

Key Stage 3 Number and Place Value Assessment

Assessment Objectives

1. Read, write, order and compare numbers up to 10 000 000 and determine the value of each digit.
2. Round to a required degree of accuracy and use rounding to estimate answers to calculations.
3. Order and compare negative numbers.
4. Solve number and practical problems using these skills.
5. Identify properties of number.

1. Write in words:

4453: _____ [1 mark]

340 096: _____ [1 mark]

1 290 432: _____ [1 mark]

9 034 023: _____ [1 mark]

2. Write in figures:

Three thousand, four hundred and eight: _____ [1 mark]

Two hundred and ninety-four thousand, one hundred and twenty-eight. _____ [1 mark]

Four million, two hundred and twenty thousand: _____ [1 mark]

A quarter of a million: _____ [1 mark]

4

4

3. What is the place value of the 2 in each of the following numbers?

198 299

_____ [1 mark]

2 040 949

_____ [1 mark]

94.542

_____ [1 mark]

3

4. Round to the nearest:

hundred: 5848

_____ [1 mark]

whole: 1.83

_____ [1 mark]

thousand: 459 900

_____ [1 mark]

2 decimal places: 0.07234

_____ [1 mark]

4

5. Use rounding to **estimate** the answers to:

$456 \times 238 =$

_____ [2 marks]

$584 \div 185 =$

_____ [2 marks]

4

6. Insert the correct sign, < or >, in between each pair of numbers:

$0 \text{ ____ } -5$ [1 mark]

$-2 \text{ ____ } 1$ [1 mark]

$-3 \text{ ____ } -10$ [1 mark]

$4 \times 2 \text{ ____ } 10 \div 2$ [1 mark]

$9 + 5 \text{ ____ } 20 - 2$ [1 mark]

$0 \times 5 \text{ ____ } 5 + 0$ [1 mark]

$32 + 15 \text{ ____ } 50 - 4$ [1 mark]

7

7. Match each number on the left with one of those on the right.

Twenty-three thousand and three.	23 300
Two hundred and thirty thousand and three.	203 030
Two hundred and three thousand and thirty.	23 003
Twenty-three thousand, three hundred.	230 030
Two hundred and thirty thousand and thirty.	230 003

[3 marks]

3

8. The table shows the local temperature at 9am every day last week. Write the days in order by their temperatures, from coldest to warmest.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
3°C	-3°C	5°C	0°C	-8°C	2°C	-1°C

[2 marks]

2

9. Use rounding to **estimate** the following:

How many days are there in 17 years?

_____ [2 marks]

I buy 18 packets of biscuits. They weigh 3070g. What is the mass of one packet of biscuits?

_____ [2 marks]

4

10. The five trees in the park are 2.23m, 3.45m, 1.61m, 2.73m and 1.98m tall. Round each height to 1 decimal place to estimate the total height of the trees in the park.

_____ [2 marks]

2

11.

125	2	4	3	16	7	1	6	64	8	11	9
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a) List all the prime numbers.

_____ [1 marks]

b) List all the triangular numbers.

_____ [1 marks]

c) List all cube numbers larger than 50.

_____ [1 marks]

d) List all the numbers which are both square numbers and cube numbers.

_____ [1 marks]

4

12. List the first 5 multiples of 7.

[1 marks]

1

13. What is the highest common factor of 24 and 32?

[1 marks]

1

14. Write 36 as a product of its prime factors.

[2 marks]

2

15. Tom has attempted to write 48 as a product of its prime factors. Help him finish his answer by filling in the missing gaps.

$$2^{\square} \times \square$$

[2 marks]

2

16. a) Complete the fourth and fifth triangular number using dots. [2 marks]

1.	2.	3.	4.	5.
				
1	3	6		

b) What will be the 10th triangular number?

_____ [1 marks]

3

End of Test

Total:

50